PrismBench and LENS: Diagnosing Evaluation Failure in Multi-Subject Personalized Generation

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Abstract

Existing text-to-image metrics are largely calibrated for single-subject alignment or global aesthetic preference, and they degrade sharply when multiple personalized subjects must be generated in the same image. In this regime, the dominant failure modes are not merely poor global semantics, but missing identities, local appearance entanglement, and incorrect interaction assignment across subjects. We argue that current evaluation practice therefore suffers from a structural blind spot: once subject count increases, many metrics return similar scores for generations that are visibly very different in identity faithfulness. To address this gap, we propose PrismBench, a benchmark for reference-conditioned multi-subject personalized generation built from paired outputs under shared prompts and shared subject references, and LENS, a diagnostic evaluator trained to predict pairwise preference together with three image-level error dimensions: Existence, Appearance, and Interaction. PrismBench combines a 60k-pair silver set generated by Nano and Mosaic and labeled by two teacher VLMs, Gemini 2.5 Flash and Gemini 3.1 Flash Lite, with a 4k-pair human evaluation set spanning six generator families. LENS is optimized with a dual-head objective that unifies pairwise ranking and diagnostic supervision, and is trained under a compute-controlled protocol across Qwen3.5 backbones from 0.8B to 4B. While the final experimental tables are completed after all runs finish, the benchmark design, filtering policy, and training protocol already define a concrete and testable route toward restoring score separability in high-subject-count personalized generation.

1 Introduction

Personalized text-to-image generation has progressed rapidly in recent years, especially in the single-subject regime where a model must preserve one reference identity while following a textual prompt. However, practical deployment increasingly requires multiple distinct reference identities to co-exist in one image, often with explicit interactions, physical contact, asymmetric role assignments, or object bindings. This multi-subject setting is substantially harder than ordinary composition because the model must preserve identity boundaries, local visual traits, and subject-to-action bindings at the same time.

The evaluation layer has not kept pace with this difficulty. In many current pipelines, quality is still measured with global similarity metrics or general preference models. These tools are useful for coarse semantic plausibility, but they are poorly matched to the real errors that dominate multi-subject personalization. A generated image can mention the right scene, contain approximately the right colors, and still be unusable because one subject is missing, two identities collapse into generic clones, clothing attributes bleed across people, or a requested interaction is assigned to the wrong subject. In such cases, a useful evaluator must preserve ranking separability even when the number of requested subjects becomes large.

This paper is motivated by a simple empirical observation: once the subject count increases, existing metrics often lose the ability to separate obviously better generations from obviously worse ones. The problem is therefore not only that current metrics are imperfect, but that they become structurally uninformative in the regime that matters most for scalable personalization. If a benchmark cannot distinguish "missing one subject" from "missing most subjects," then it cannot support real model development or credible ablation.

We formalize this failure mode through two coupled contributions. First, we introduce **PrismBench**, a benchmark specifically designed for reference-conditioned multi-subject generation. Each task consists of a prompt, a set of reference subjects, and a paired comparison between two generated images produced under the same prompt and references. The current benchmark construction uses a 60k-pair silver pipeline built from Nano and Mosaic generations labeled by Gemini 2.5 Flash and Gemini 3.1 Flash Lite, together with a 4k-pair human evaluation set spanning Nano, Mosaic, Flux, Seedream 4.5, GPT-Image-1.5, and GLM. Second, we propose **LENS**, a learned evaluator that takes the references, prompt, and generated image pair as input, then predicts both pairwise preference and three interpretable image-level diagnostic dimensions: *Existence*, *Appearance*, and *Interaction*.

The design choices in PrismBench and LENS are driven by a practical principle: preserve maximal ranking signal while minimizing annotation burden. Instead of requiring annotators to label the exact number of missing subjects or to score each subject with fine-grained severity levels, we adopt binary image-level diagnosis and a simple A/B preference decision. This makes large-scale annotation feasible, improves consistency, and still enables the score head to learn a continuous severity scale from pairwise comparisons. The silver pipeline is intentionally conservative: preference conflicts are dropped, while diagnostic disagreement is retained as averaged soft supervision.

The expected outcome is not merely another scalar metric, but a benchmark-and-metric stack that can answer three questions at once: which image is better, why it is better, and whether the metric remains discriminative when subject count and interaction difficulty increase. The paper is therefore centered on a broader thesis: *multi-subject personalized generation requires a new evaluation protocol, not just a stronger backbone.* The experimental program then becomes straightforward: train LENS on the large silver set, stress-test it across backbone scales and update modes, and compare it against standard baselines on a broader human evaluation set collected from multiple generator families.

The remainder of the paper is organized as follows. Section 2 reviews the limitations of existing evaluation paradigms. Section 3 defines the problem setup and diagnostic taxonomy. Section 4 introduces PrismBench, including data collection and annotation protocols. Section 5 presents the LENS evaluator. Section 6 describes the training pipeline. Section 7 outlines the planned experiments and success criteria, and Section 8 concludes with discussion and limitations.

2 Related Work

2.1 Text-to-Image Evaluation

Most current image generation metrics were not designed for reference-conditioned multi-subject identity preservation. Global embedding metrics such as CLIP [6] and DINOv2 [5] are effective when the dominant question is whether an image matches a caption at a coarse semantic level, but they tend to average away localized identity failures. General reward and preference models, such as ImageReward [7] and PickScore [4], can provide useful aesthetic or holistic judgments, yet they usually remain black-box signals and often prioritize visual appeal over strict identity fidelity. Our work focuses on the regime where these shortcomings become critical: multiple reference identities, non-trivial interactions, and identity-preserving generation under the same prompt.

2.2 Compositional and VQA-Style Benchmarks

Prior composition-oriented benchmarks like T2I-CompBench [3] demonstrate that text-to-image models struggle when multiple attributes or spatial relations must be satisfied at once. However, most of these settings assume text-only specification rather than explicit reference subjects. This distinction matters. In multi-subject personalization, the task is not only to realize a caption, but to correctly bind the right visual identity to the right local attributes and interactions. A benchmark that ignores reference images cannot fully evaluate identity collapse, clone failure, or cross-subject appearance leakage.

2.3 Subject-Driven and Personalized Generation

Subject-driven generation methods are typically evaluated with single-subject identity preservation metrics such as ArcFace [2] or limited qualitative examples. The multi-subject regime introduces a different failure geometry, as noted in our prior work on multi-subject evaluation [1]. Identity preservation must scale with subject count, and the evaluator must remain discriminative as the scene grows denser. This gap motivates a benchmark that directly stress-tests scalability rather than treating multi-subject generation as a minor extension of the single-subject case.

2.4 Learned Evaluators and Diagnostic Metrics

Learned metrics are especially promising when existing handcrafted or global embedding metrics fail to reflect human judgment. Our work follows this spirit but differs in two important ways. First, LENS is explicitly reference-conditioned and pairwise trained under shared prompts. Second, it produces both a scalar preference signal and interpretable diagnostic dimensions. This makes the metric useful not only for ranking models, but also for diagnosing why they fail.

3 Problem Setup and Taxonomy

3.1 Reference-Conditioned Multi-Subject Evaluation

Each evaluation task contains a text prompt, a set of N reference subjects, and two generated images produced under the same prompt and references. The goal of the metric is to determine which generated image is better aligned with the full task specification. Unlike standard text-to-image evaluation, the metric must verify not only prompt plausibility but also identity-specific preservation across multiple subjects.

3.2 Why Global Metrics Fail

In high-subject-count settings, a model may partially satisfy the prompt while violating the core identity constraints that matter to users. For example, a scene may contain the correct number of figures only approximately, merge several identities into generic lookalikes, or swap actions and props across subjects. Global similarity metrics compress these distinct outcomes into one embedding comparison, which often erases the local evidence needed to preserve ranking separability.

3.3 Three Diagnostic Dimensions

We decompose multi-subject failure into three orthogonal image-level dimensions:

- **Existence**: whether all requested reference subjects are present without omission, cloning, or severe unidentifiable collapse.
- **Appearance**: whether the generated image preserves correct physical structure and local appearance for the requested reference subjects, without severe distortion or cross-subject feature bleeding.
- **Interaction**: whether subject-to-subject relations, actions, and prop assignments are aligned with the prompt.

Each dimension is annotated as a binary image-level variable. A score of 1 indicates no observed failure on that dimension, while 0 indicates that the image fails the dimension in a noticeable way. This choice is intentionally minimalist: it reduces ambiguity for annotators while still supplying a strong diagnostic regularizer during training.

3.4 Preference as a Continuous Severity Proxy

Binary diagnosis alone cannot distinguish all severity levels. In particular, an image missing one subject and an image missing several subjects may both receive Existence$= 0$. We address this by treating pairwise preference as the channel through which severity becomes learnable. The diagnostic head provides categorical structure, while the ranking head learns a continuous latent score that places more severe failures lower whenever annotators systematically prefer the less damaged image. This combination preserves annotation simplicity without discarding score granularity.

4 PrismBench

4.1 Dataset Overview

PrismBench is designed as a stress-test benchmark for multi-subject personalized generation. The benchmark spans subject counts $N \in \{2, 4, 6, 8\}$ and is built around controlled pairwise comparison: for each task, the same prompt and the same reference subjects are paired with two generated candidates, denoted Image A and Image B. This setup suppresses nuisance variance from unrelated scene changes and turns the evaluation target into a clean ranking question under fixed task conditions.

4.2 Hierarchical Prompt Construction

To ensure rigorous control over scene complexity and interaction types, we construct a 60K prompt set for multi-entity image generation. The construction process is strictly hierarchical. We generate 60K prompts from 15K distinct seeds. Each seed produces four related prompts at complexity levels $N \in \{8, 6, 4, 2\}$. We first construct the most complex prompt ($N = 8$) and then obtain lower-complexity versions by systematically removing entities and their corresponding relations. This ensures that prompt complexity decreases in a controlled and consistent way across levels, allowing us to isolate the exact point at which a generator or metric fails.

All prompts describe a scene in a white studio to minimize background distractors. To reduce unrealistic size relationships during generation, we prepend a global constraint to every prompt: *"White studio. Keep all entities at realistic scale."*

4.3 Reference Pool and Factorized Control

The prompts are grounded in a fixed pool of 80 reference images (resized to 512×512), consisting of 30 human references and 50 non-human references (animals and objects). The prompt space is factorized across two key axes:

- **Relation Category**: Prompts are assigned to one of three structural types: `no_interaction_no_occlusion` (strictly non-contact), `occlusion_no_interaction` (spatial overlap without active engagement), and `occlusion_interaction` (explicit physical verbs, excluding weak descriptions like gaze).
- **Human/Object Ratio**: Each prompt follows a strict composition ratio: `balanced`, `human_heavy`, or `object_heavy`. The exact distribution of human (H) and non-human (N) entities across the four complexity levels is detailed in Table 1.

Table 1: Number of human (H) and non-human (N) entities at each complexity level for the three ratio types.

Ratio Type	Level 8	Level 6	Level 4	Level 2
balanced	4H, 4N	3H, 3N	2H, 2N	1H, 1N
human_heavy	6H, 2N	4H, 2N	3H, 1N	2H, 0N
object_heavy	2H, 6N	2H, 4N	1H, 3N	0H, 2N

This creates $3 \times 3 \times 4 = 36$ distinct buckets, across which the 60,000 prompts are distributed approximately uniformly.

4.4 Quality Control for Prompt Generation

To guarantee that the text prompt accurately reflects the intended combinatorial structure, we enforce strict automatic quality checks during construction: (1) every assigned entity must be explicitly mentioned; (2) no unassigned entities may appear; (3) the sum of humans and non-humans must match the target level; (4) lower-level prompts must remain strict subsets of higher-level prompts from the same seed; (5) exact duplicate prompts are removed; (6) a single subject is not reused as the active subject of multiple independent sentences. The final finalized prompt set guarantees 100% entity coverage and strict $8 \rightarrow 6 \rightarrow 4 \rightarrow 2$ hierarchical consistency. The broader goal is to build an evaluation stack that supports both large-scale metric training and generator-agnostic meta-evaluation.

4.5 Controlled Paired Generation

The benchmark uses two generated images under identical task conditions. Because both images share the same prompt and subject references, differences in the final labels are more likely to reflect subject-level alignment quality rather than unrelated background content. This paired design also makes the ranking supervision natural: annotators are asked which image is better for the same task, not which image is absolutely good in isolation.

4.6 Scale

The current benchmark construction has two layers:

- **Silver training set**: roughly 60k candidate pairs generated by Nano and Mosaic and labeled by two teacher VLMs, Gemini 2.5 Flash and Gemini 3.1 Flash Lite.
- **Human evaluation set**: roughly 4k pairs curated for paper-facing evaluation. Among them, 1.5k pairs come from Nano and Mosaic, while the remaining 2.5k pairs come from `flux2_klein_9b_kv`, Seedream 4.5, GPT-Image-1.5, and GLM.

After filtering, the current silver manifest contains 57,055 usable pairs, split into 51,328 training pairs, 2,847 validation pairs, and 2,880 test pairs. These splits are performed by grouped `seed_id` rather than individual samples so that closely related prompt families do not leak across train, validation, and test. The human evaluation set is intentionally broader in generator coverage than the silver training set, allowing the final meta-evaluation to test whether a metric trained on Nano and Mosaic transfers beyond those two generators.

4.7 Label Schema

Each stored sample contains the prompt, the reference subject list, the path to Image A and Image B, and diagnostic labels derived from two teacher sources. The stored target consists of:

- a binary preference decision between Image A and Image B,
- a three-dimensional binary diagnostic vector for Image A,
- a three-dimensional binary diagnostic vector for Image B.

The schema intentionally remains image-level. We do not require the labeling pipeline to count the exact number of missing subjects or assign per-subject severity levels. This choice improves label stability and allows the model to learn severity through pairwise ranking instead of relying on brittle fine-grained scalar annotations.

4.8 Silver Filtering Rules

The current `v2` build process applies conservative filtering before a pair is admitted to the final manifests:

- if the two teacher label sources disagree on the pairwise preference, the sample is dropped;
- if the diagnostic labels differ, the three category scores are averaged across the two sources;
- if any required reference image is missing, the sample is dropped;
- if either generated image is missing, the sample is dropped.

This policy is intentionally asymmetric. Preference disagreement destroys the ranking target and is therefore treated as fatal noise, whereas diagnostic disagreement is treated as soft uncertainty that can still provide useful supervision after averaging. In other words, the build process prefers a smaller but cleaner ranking set over a larger set with unstable ordering signal.

4.9 Annotation Protocol

The current training pipeline uses dual-teacher silver labels under a shared schema. Two teacher VLM outputs are collected for each task, then merged only if the ranking decision is consistent. The human evaluation layer is separated from this training pipeline and is designed for paper-facing metric comparison across a wider set of generators. This distinction matters: the silver set provides scale, while the human set provides stronger evidence of alignment and transfer.

4.10 Why PrismBench Matters

PrismBench is not simply another paired preference set. Its core purpose is to isolate the exact evaluation regime where existing metrics fail: shared prompt, shared references, multiple requested subjects, and controlled comparison between two generated images. This is the setting required to measure whether a metric still has discriminative power after identity complexity grows.

5 LENS

5.1 Design Goal

LENS is designed to restore metric separability in multi-subject personalized generation. It is not sufficient for a metric to output lower scores as subject count increases. A useful evaluator must still distinguish obviously better generations from obviously worse ones under the same prompt-reference condition, especially at high subject counts where conventional metrics collapse.

5.2 Model Interface

LENS takes as input the reference subjects, the prompt, and a generated image pair. The references define the identity targets; the prompt defines the semantic and interaction constraints; the two generated images provide the paired comparison for ranking. The output contains:

- a scalar preference score for each image,
- a binary diagnostic vector for each image over Existence, Appearance, and Interaction.

5.3 Architecture

The evaluator uses a multimodal backbone and attaches two lightweight heads on top of the shared representation:

- a **score head** that outputs a scalar score,
- a **classification head** that outputs the three diagnostic logits.

This dual-head design is important. The score head captures fine-grained ordering information that binary class labels cannot represent, while the classification head regularizes the shared representation toward interpretable failure modes.

5.4 Why the Binary Taxonomy is Enough

At first glance, a binary diagnostic scheme may appear too coarse for nuanced quality assessment. However, LENS does not rely on the diagnostic head alone. The role of the diagnostic head is to define *what kind of error is present*. The role of the score head is to define *how bad the image is relative to another image*. This separation allows the training objective to remain simple for annotators while preserving continuous score discrimination in the learned metric.

5.5 Expected Failure Sensitivity

The main expected advantage of LENS over global metrics is its ability to remain reference-aware and locally diagnostic when N becomes large. If one generator preserves most identities while another collapses several of them, LENS should assign a higher preference score to the first image even when conventional metrics report nearly overlapping values. This property is the core criterion for success in the final evaluation.

6 Training Protocol

6.1 Ranking Objective

For each task, the model receives Image A and Image B generated under the same prompt and the same reference subjects. The scalar outputs of the score head are optimized with a pairwise ranking loss so that the preferred image receives a higher score than the rejected image. This setup directly matches the final use case of the metric: relative comparison under fixed task conditions.

6.2 Diagnostic Objective

In parallel, the classification head is trained with binary cross-entropy over the three diagnostic dimensions for both images. This auxiliary supervision is critical because it prevents the shared representation from collapsing into a purely aesthetic ranking signal. Instead, the model is pushed to attend to concrete image-level failure modes that are meaningful to users and reviewers.

6.3 Multi-Task Loss

The overall training loss is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{rank}}\mathcal{L}_{\text{rank}} + \lambda_{\text{cls}}\mathcal{L}_{\text{cls}}.$$

The ranking term teaches global ordering, while the classification term supplies interpretable structure. In the current implementation, the default weights are $\lambda_{\text{rank}} = 1.0$ and $\lambda_{\text{cls}} = 0.5$, which biases optimization toward preserving ranking quality while still forcing the shared representation to encode diagnostic cues.

6.4 Backbone Family and Training Modes

The current paper-facing training pipeline instantiates LENS on top of Qwen3.5-VL backbones using Unsloth acceleration. We run three backbone scales, `0.8B`, `2B`, and `4B`, and two parameter-update modes:

- **layer_only**: unfreeze the last transformer layers and train the diagnostic heads;
- **lora_layer**: attach LoRA adapters while also unfreezing the last transformer layers.

This produces six paper-facing training configurations in total. The design goal is not to search for a separate hyperparameter optimum for every backbone, but to keep the comparison protocol reviewer-friendly and compute-balanced across scales and modes.

6.5 Why This Should Recover Score Separability

The training design is explicitly targeted at the failure mode of existing metrics. Global similarity methods lose separability because localized identity errors are averaged into a coarse representation. LENS instead receives repeated supervision that says: under identical prompts and references, one image is better, and the reason is linked to Existence, Appearance, or Interaction. Over many pairs, this teaches the model to build a score surface aligned with human preference in the high-subject regime.

6.6 Data Filtering

The silver pipeline applies several quality filters before training. Preference conflicts across the two teacher sources are removed entirely, because they destroy the ordering target required by the margin-ranking objective. For the three diagnostic dimensions, the two teacher outputs are averaged rather than thresholded away, which converts mild disagreement into soft supervision. Samples are also removed if any required reference image or either generated candidate image is missing. This filtering stage is necessary because pairwise ranking models are especially sensitive to noisy preference labels. A slightly smaller but cleaner silver set is preferable to a larger set with ambiguous ordering signal.

6.7 Implementation-Level Training Budget

The current training protocol is intentionally compute-controlled. All six main model configurations are trained with a shared fixed optimizer-step budget of 600 updates. The physical batch size is 1 due to multimodal memory constraints, while gradient accumulation is auto-scaled to reach an effective batch size of 16. Additional default settings are shared across runs: learning rate 2×10^{-5}, weight decay 0.01, warmup ratio 0.03, seed 3407, gradient clipping 1.0, and early stopping patience 1. This protocol is designed to support fair ablation under limited wall-clock budget, not to claim that 600 steps fully saturate every backbone.

6.8 Current Data Split Used For Training

The present 'v2' manifests contain 57,055 usable pairs after filtering, split into 51,328 train, 2,847 validation, and 2,880 test examples. Training uses the train split, checkpoint selection uses the validation split, and the test split is reserved for final reporting. In the current large-scale run plan, the main six backbone-mode combinations are first trained under a shared budget with validation monitoring; full test-table reporting is completed afterward once all runs finish successfully.

6.9 Validation Strategy

The current codebase performs model selection on the validation manifest and saves both epoch checkpoints and the best checkpoint under the monitored validation loss. For the full paper, we keep two evaluation levels conceptually separate: validation on the current 'v2' split for training-time model selection, and final human-facing meta-evaluation on a golden test set once that set is finalized. This preserves the distinction between teacher-derived supervision used for optimization and stronger human-grounded evidence used for the final claim.

7 Experiments

7.1 Experimental Questions

The empirical section is organized around four questions:

1. Do existing evaluators lose discriminative power as subject count increases?
2. Does LENS better align with pairwise human preference than standard image-text or image-image metrics?
3. Does multi-task diagnostic supervision improve over ranking-only training?
4. How stable are the conclusions across backbone scales and update modes?

7.2 Ready-To-Run Experimental Setup

The current experiment plan is already fixed even though the final tables are still being collected. We train six LENS variants using Qwen3.5 backbones at `0.8B`, `2B`, and `4B` scales, each under `layer_only` and `lora_layer` modes. All six runs use the same paper-facing budget of 600 optimizer steps. This shared-budget design is intended to make the main comparison reviewer-friendly under a realistic compute constraint. Training is performed on the 60k-pair dual-teacher silver pipeline, while paper-facing comparison is centered on the broader 4k-pair human evaluation set.

7.3 Baselines

The current baseline panel for metric comparison includes the evaluators that are already integrated into our analysis workflow: CLIP-T, CLIP-I, DINOv2, ArcFace, ImageReward, SCR, and REFVNL. These methods were selected because together they cover text-image similarity, image-image similarity, identity-oriented features, reward-model style preference signals, and specialized reference-aware scoring. In the final paper version, these baselines will be evaluated on the same multi-generator human set used to assess LENS.

7.4 What Can Be Written Now

Even before the final result tables are complete, the following parts of the section are ready to write and should remain in the paper draft:

- the evaluation questions and success criteria;
- the fixed training-budget protocol for the six LENS variants;
- the distinction between the 60k-pair silver training set and the 4k-pair human evaluation set;
- the baseline list and why each baseline is included;
- the planned breakdown by subject count $N \in \{2, 4, 6, 8\}$;
- the list of ablations and qualitative analyses to report.

7.5 Evaluation Data Regime

Our evaluation design intentionally separates scale from trustworthiness. The silver set supplies the large number of pairwise comparisons needed to train a reference-conditioned evaluator, but the central empirical claim of the paper is assessed on human judgments. The 4k-pair human evaluation set also spans more generator families than the training set, which makes it possible to test cross-generator transfer rather than only in-domain agreement on Nano and Mosaic.

7.6 What Should Stay Empty For Now

The following subsections should remain explicit placeholders until the full runs and evaluation outputs are finalized:

- **Main result table**: final metric-to-human alignment numbers;
- **Subject-count scaling**: plots or tables showing score separability as N increases;
- **Backbone comparison**: the final comparison across `0.8B`, `2B`, and `4B`;
- **Mode comparison**: the final comparison between `layer_only` and `lora_layer`;
- **Qualitative case studies**: curated failure and success examples with figures.

7.7 Planned Main Results Table

Table 2 is reserved for the final paper-facing summary once all runs finish.

7.8 Planned Ablations and Analyses

Once the runs and evaluation files are complete, we will populate this section with:

- **Meta-evaluation against human judgment**: Kendall-τ, Spearman, and pairwise accuracy on the held-out human set.
- **High-subject-count separability**: score-gap and overlap analysis between stronger and weaker generations for $N \in \{2, 4, 6, 8\}$.
- **Cross-generator transfer**: comparison between generators seen in silver training data and generators appearing only in human evaluation.
- **Ablations**: ranking-only versus multi-task training, with and without agreement-based filtering.

Table 2: Placeholder for the main result table. Fill after all evaluation outputs are finalized.

Method	Pair Acc.	Spearman	Kendall-τ	Notes
CLIP-T	–	–	–	text-image baseline
CLIP-I	–	–	–	image-image baseline
DINOv2	–	–	–	self-supervised visual baseline
ArcFace	–	–	–	identity baseline
ImageReward	–	–	–	reward-model baseline
SCR	–	–	–	subject completeness baseline
REFVNL	–	–	–	reference-aware baseline
LENS (0.8B, layer_only)	–	–	–	pending
LENS (0.8B, lora_layer)	–	–	–	pending
LENS (2B, layer_only)	–	–	–	pending
LENS (2B, lora_layer)	–	–	–	pending
LENS (4B, layer_only)	–	–	–	pending
LENS (4B, lora_layer)	–	–	–	pending

- **Qualitative case studies**: examples where global metrics collapse but LENS preserves the correct ordering.

The central experimental claim to validate remains the same: existing metrics lose score separability as subject count increases, whereas a reference-conditioned evaluator trained with pairwise ranking and diagnostic supervision can remain more discriminative and better aligned with human preference.

8 Discussion and Conclusion

8.1 Why This Paper Matters

The broader contribution of this work is to reframe multi-subject personalized generation as an evaluation problem rather than only a generation problem. If the community cannot reliably measure subject omission, appearance entanglement, and interaction errors, progress in model design will remain difficult to interpret. PrismBench and LENS aim to provide the missing measurement infrastructure.

8.2 Current Limitations

The proposed benchmark and metric design still involve several limitations. First, to establish a fair and rigorously controlled ablation across various backbone sizes (0.8B to 4B) and fine-tuning paradigms (layer-only vs. LoRA), the current LENS models were trained under a strict compute-constrained budget (600 optimizer steps, observing roughly 9.6K samples). However, PrismBench provides up to 60K full silver training pairs. Second, the silver set depends on teacher VLMs, so any systematic teacher bias may propagate into training even after agreement filtering. Third, the diagnostic labels are intentionally coarse and operate at the image level rather than the per-subject level. This is a deliberate trade-off for annotation stability, but it may limit the granularity of downstream analysis. Fourth, the current training set is narrower in generator coverage than the human evaluation set, so strong transfer on unseen generators must be demonstrated empirically rather than assumed. Fifth, the current setup focuses on pairwise evaluation under controlled prompts and reference subjects; broader generalization to arbitrary open-world scenes remains to be studied.

8.3 Future Work

Several next steps follow naturally. A critical direction is to explore the *Scaling Laws* of multi-subject evaluators: investigating the absolute performance ceiling of LENS when fully trained on the entire 60K dataset without compute restrictions. Another direction is to study whether the learned evaluator can generalize across unseen generators without retraining. A third is to extend the benchmark toward video or temporally consistent multi-subject generation. A fourth is to investigate more structured diagnostic heads once sufficient evidence is available that the simpler binary scheme has saturated.

8.4 Conclusion

We presented the conceptual and methodological foundation for PrismBench and LENS, a benchmark-and-metric framework aimed at evaluating multi-subject personalized generation under the failure modes that matter most in practice. The benchmark emphasizes controlled pairwise comparison, a large dual-teacher silver training pipeline, a broader multi-generator human evaluation layer, and image-level diagnostic supervision. The metric couples pairwise ranking with interpretable error prediction in order to preserve score separability as subject count increases. With the experimental section completed, this framework is positioned to test the central claim of the paper: that existing metrics collapse in the high-subject regime, while a reference-conditioned evaluator can remain discriminative and human-aligned.

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